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#include <stdio.h>
#include <stdlib.h>

struct Node {
    int info;
    struct Node* prev;
    struct Node* next;
};

struct Node* createNode(int info) {
    struct Node* p = (struct Node*)malloc(sizeof(struct Node));
    p->info = info;
    p->prev = NULL;
    p->next = NULL;
    return p;}

void insertEnd(struct Node* start, int info) {
    struct Node* p = createNode(info);
    if (start == NULL) {
        start = p;
        return;}
    struct Node* temp = start;
    while (temp->next != NULL) {
        temp = temp->next; }
    temp->next = p;
    p->prev = temp; }

struct Node* insertLeft(struct Node* start, int item, int info) {
    struct Node* temp = start;
    while (temp != NULL && temp->info != item) {
        temp = temp->next; }
    if (temp == NULL) {
        printf("Item %d not found!\n", item);
        return start;}
    struct Node* p = createNode(info);
    p->next = temp;
    p->prev = temp->prev;
    if (temp->prev != NULL) {

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    struct Node* prev = temp->prev;

    prev->next = p;
} else {
    start = p;}

temp->prev = p;
return start; }

struct Node* deleteNode(struct Node* start, int item) {
    struct Node* temp = start;

    while (temp != NULL && temp->info != item) {
        temp = temp->next;}

    if (temp == NULL) {
        printf("Node with value %d not found!\n", item);
        return start; }

    if (temp->prev != NULL) {
        struct Node* prev = temp->prev;
        prev->next = temp->next;
    } else {
        start = temp->next; }

    if (temp->next != NULL) {
        struct Node* next = temp->next;
        next->prev = temp->prev;}

    free(temp);

    printf("Node with value %d deleted.\n", item);
    return start; }

void display(struct Node* start) {
    struct Node* temp = start;
    printf("List: ");
    while (temp != NULL) {
        printf("%d ", temp->info);
        temp = temp->next;}
    printf("\n"); }

int main() {
    struct Node* start = NULL;
    int choice, info, item;

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while (1) {
    printf("\nMenu:\n");
    printf("1. Insert at end\n");
    printf("2. Insert to the left of a node\n");
    printf("3. Delete a node by value\n");
    printf("4. Display list\n");
    printf("5. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    if (choice == 1) {
        printf("Enter value to insert: ");
        scanf("%d", &info);
        if (start == NULL) {
            start = createNode(info);}
        else {
            insertEnd(start, info); }
    } else if (choice == 2) {
        printf("Enter item (node value to insert left of): ");
        scanf("%d", &item);
        printf("Enter info for new node: ");
        scanf("%d", &info);
        start = insertLeft(start, item, info);
    } else if (choice == 3) {
        printf("Enter value of node to delete: ");
        scanf("%d", &item);
        start = deleteNode(start, item);
    } else if (choice == 4) {
        display(start);
    } else if (choice == 5) {
        break;
    } else {
        printf("Invalid choice!\n");
        } }
return 0; }

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OUTPUT:

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 1

Enter value to insert: 10

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 1

Enter value to insert: 20

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 1

Enter value to insert: 30

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 1

Enter value to insert: 40

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 2

Enter item (node value to insert left of): 20

Enter info for new node: 15

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 2

Enter item (node value to insert left of): 10

Enter info for new node: 79

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 3

Enter value of node to delete: 40

Node with value 40 deleted.

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 4

List: 79 10 15 20 30

Menu:

1. Insert at end
2. Insert to the left of a node
3. Delete a node by value
4. Display list
5. Exit

Enter your choice: 5

Process returned 0 (0x0) execution time : 48.561 s

Press any key to continue.

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